

[ML-FIELD-GUIDE]

PROJECT NO. ML-2026 | STATUS: REFERENCE

MACHINE LEARNING

A Technical Field Guide to Algorithms, Architectures, and
Modern Practice

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WHAT IS MACHINE LEARNING?

FIG. 01 – DEFINITION & SCOPE

SPEC: CORE-01
STATUS: ACTIVE

Systems that learn patterns from data without explicit programming

A branch of AI focused on building predictive models

Enables computers to improve at tasks through experience

Three core components: data, model, and training algorithm

DIAGRAM 1.0 – CORE COMPONENTS

01

DATA

Raw observations for pattern extraction

02

MODEL

Mathematical structure encoding learned patterns

03

ALGORITHM

Optimization procedure that trains the model

TYPES OF MACHINE LEARNING

FIG. 02 – TAXONOMY

SUPERVISED

Supervised Learning

Labeled training data – predicts outcomes from known input-output pairs. The model learns a mapping function.

UNSUPERVISED

Unsupervised Learning

Finds hidden patterns in unlabeled data. Discovers intrinsic structure without predefined categories.

REINFORCEMENT

Reinforcement Learning

Agent learns via rewards and penalties. Optimizes behavior through interaction with an environment.

SELF-SUPERVISED

Self-Supervised Learning

Generates labels from raw data itself. Foundation of modern large language and vision models.

SUPERVISED LEARNING

FIG. 03 – METHODS & APPLICATIONS

TASK TYPES

Classification

Assigns input to discrete categories.
Output is a class label.

Regression

Predicts continuous numeric values.
Output is a real number.

KEY ALGORITHMS

- Linear Regression
- Decision Trees
- Support Vector Machines
- k-Nearest Neighbors
- Random Forests

APPLICATIONS

- Spam detection
- Price prediction
- Medical diagnosis
- Credit scoring
- Image classification

NEURAL NETWORKS & DEEP LEARNING

FIG. 04 - ARCHITECTURE OVERVIEW

Inspired by the structure of biological neurons – interconnected nodes that process and transmit signals

Layers: Input → Hidden → Output – each layer learns increasingly abstract representations

Deep networks learn hierarchical feature representations – from edges to shapes to objects

CORE ARCHITECTURES

CNN

RNN

Transformer

GAN

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Output

UNSUPERVISED LEARNING

FIG. 05 — METHODS TAXONOMY

SPEC: UL-01
NO LABELS REQUIRED

CLUSTERING

- ▶ k-Means
- ▶ DBSCAN
- ▶ Hierarchical Clustering

DIMENSIONALITY REDUCTION

- ▶ PCA
- ▶ t-SNE
- ▶ UMAP

ANOMALY DETECTION

- ▶ Isolation Forest
- ▶ Autoencoders

GENERATIVE MODELS

- ▶ GANs — Generative Adversarial Networks
- ▶ VAEs — Variational Autoencoders

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MODEL TRAINING & OPTIMIZATION

FIG. 06 – TRAINING PIPELINE

SPEC: OPT-01
STATUS: CRITICAL PATH

Loss Function

Measures prediction error. Guides the optimizer by quantifying how wrong the model is.

Gradient Descent

Minimizes the loss iteratively. Updates weights in the direction that reduces error.

Bias-Variance Tradeoff

Overfitting vs. underfitting – finding the sweet spot between memorization and generalization.

REGULARIZATION TECHNIQUES

L1

Lasso – promotes sparsity by driving weights to zero

L2

Ridge – penalizes large weights, smoother solutions

Dropout

Randomly disables neurons during training to prevent co-adaptation

EVALUATION METRICS

FIG. 07 – PERFORMANCE MEASUREMENT

DATA SET: BENCHMARK
UNITS: METRIC

CLASSIFICATION

Accuracy

Correct predictions / total

Precision

TP / (TP + FP)

Recall

TP / (TP + FN)

F1 / AUC-ROC

Harmonic mean / area under curve

REGRESSION

MAE

Mean Absolute Error

MSE / RMSE

Mean Squared / Root Mean Squared Error

R-squared

Proportion of variance explained

VALIDATION STRATEGY

SPLIT PROTOCOL

Train – 70%

Validation – 15%

Test – 15%

Cross-validation provides robust performance estimates across multiple folds

ML TOOLS & FRAMEWORKS

FIG. 08 – ECOSYSTEM OVERVIEW

REV: CURRENT
STATUS: PRODUCTION

LANGUAGES

Py

Python

R

R Language

Jl

Julia

ML LIBRARIES

scikit-learn

TensorFlow

PyTorch

Keras

DATA

Pandas

NumPy

Apache Spark

MLOps

MLflow

Weights & Biases

Kubeflow

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IMPLEMENTATION PHASE: ONGOING

STATUS: FORWARD-LOOKING

PRIORITY: HIGH

THE ROAD AHEAD

Foundation models and large-scale pretraining

Responsible AI: fairness, explainability, safety

Multimodal learning: vision, language, audio

Edge ML: on-device inference for real-time applications

SECT. 10

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